

Cover Letter

Dear Editor,

We would like to submit our manuscript entitled “Timestep-Conditioned Adapter Scaling for Multi-view Diffusion Texture Generation” for consideration for publication in *Computers & Graphics*, as part of the Virtual Special Issue CAG_SS_CAD/Graphics 2026. The manuscript is original, has not been published previously, and is not under consideration elsewhere. This submission follows the editor’s decision on our previous submission of the same manuscript (CAG-D-26-00954); the manuscript has been thoroughly revised in response to all editorial and reviewer comments, a point-by-point response is provided at the beginning of the manuscript PDF, and a marked-up copy is included for the editor’s convenience.

In this manuscript, we study a fine-grained inference-time control problem in geometry-controlled multi-view diffusion texturing: how strongly, and at which denoising stages, a geometric adapter should act. We first show that a uniform adapter scale is a blunt control — aggressive uniform scales improve structural metrics such as FG-SSIM but suppress the texture synthesis capability of the base model, causing surface flattening and loss of high-frequency detail. We then formalize the effect with a measurable Correction–Alignment–Interference (CAI) stage-utility model and derive Timestep-Conditioned Adapter Scaling (TCAS), a training-free low–high–low schedule that concentrates geometric correction in the middle denoising stage. The schedule $C3 = (1.25, 2.50, 1.25)$ is selected using only a 24-object probe set and then evaluated unchanged, without further search, on the strictly disjoint 276-object holdout (pooled 300-object statistics also reported), where it matches the structure of aggressive scaling while improving PSNR by 0.96 dB, retaining substantially more texture, and receiving the highest overall preference (58.1%) in a blinded human study with cluster-aware significance analysis. To the best of our knowledge, no prior work analyzes the shape–texture trade-off of adapter scaling or derives a timestep-conditioned schedule from measurable stage utilities; the closest prior art (MVPainter, arXiv 2025; MV-Adapter, ICCV 2025; ControlNet, ICCV 2023; limited-interval classifier-free guidance, NeurIPS 2024) is cited and compared explicitly in Sections 2.3 and 4.4 of the manuscript.

The reference list has been carefully verified: citations that previously existed only as preprints have been updated to their peer-reviewed versions where available (e.g., TEXTure at SIGGRAPH 2023, Text2Tex at ICCV 2023, SyncDreamer at ICLR 2024, Wonder3D at CVPR 2024, Era3D at NeurIPS 2024, T2I-Adapter at AAAI 2024, DoRA at ICML 2024, MV-Adapter at ICCV 2025), genuinely unpublished works retain their arXiv identifiers, and recent relevant literature from *Computers & Graphics*, *IEEE TVCG*, *Computer Graphics Forum*, SIGGRAPH Asia, and *The Visual Computer* is now cited and discussed. For transparency, a copy of the original conference submission (CAD/Graphics 2026, Paper 75) is provided as a separate supplementary file.

Thank you very much for your attention and consideration.

Sincerely,

The Authors

(Per the journal's double-anonymized review policy, this cover letter contains no author details; author names, affiliations, and the corresponding author's contact information are provided on the separately uploaded title page and in Editorial Manager.)